

ECON 0100 | Demo B

Demos are similar to Checkpoints, often taken directly from past semesters. Work through the problems and check your work against mine. Practice answering clearly and completely. Show your work so someone else can understand your thought process. You are encouraged to work in small groups. Find a study room, grab some classmates, and work together on a whiteboard.

Dragontear

Dragontear is a rare and powerful substance used in potions, spells, and fire-breathing, harvested from dragons while they laugh uncontrollably. The supply and demand equations for dragontear are known to be the following:

$$P_d = 60 - \frac{1}{10}Q_d \qquad P_s = \frac{1}{5}Q_s$$

Prices are in coins and quantity is in vials. Use a graph to guide your work.

Q1 | Demand and Consumer Surplus (B1.1, B1.2)

Use a graph to plot this demand curve, and find the quantity demanded at a price of 20, 30, and 40.

a) Quantity demanded at a price of 20 _____, at 30 _____, at 40 _____

b) What is the marginal benefit of the 300th vial? _____

Find and label the consumer surplus at a price of 30. On the same graph, show what happens to consumer surplus when the price increases to 40.

c) CS at 30 _____ CS at 40 _____

Q2 | Supply and Producer Surplus (B2.1, B2.2)

Use a graph to plot this supply curve, and find the quantity supplied at a price of 20, 30, and 40.

a) Quantity supplied at 20 _____, at 30 _____, at 40 _____

b) What is the marginal cost of the 100th vial? _____

Find and label the producer surplus at a price of 30. On the same graph, show what happens to producer surplus when the price increases to 40.

c) PS at 30 _____ PS at 40 _____

Q3 | Equilibrium for Dragontear (B3.1, B4.1)

Use a graph to describe the market for Dragontear:

a) Calculate equilibrium price: _____ Find equilibrium quantity: _____

- b) Calculate consumer surplus: _____ Calculate producer surplus: _____
- c) At a price of 30, is there a shortage or an excess, and how large? _____
- d) What is the deadweight loss if only 100 vials are exchanged? _____

Q4 | Price Floor (B4.2)

Use a graph of the demand curve and the supply curve to evaluate the welfare effects of a price floor of 45.

- a) CS _____ Change in CS from Equilibrium _____
- b) PS _____ Change in PS from Equilibrium _____
- c) DWL _____ Shortage/Excess _____

Q5 | Second Wizarding War (B5.1, B5.3)

The second wizarding war limited the population of dragons. Use a graph to aid your discussion of the impact the war had on the market for dragon-tear.

- a) The demand curve: *shifted in, stayed the same, shifted out* b) The supply curve: *shifted in, stayed the same, shifted out*
- c) The equilibrium price: *increased, stayed the same, decreased* d) The equilibrium quantity: *increased, stayed the same, decreased*

In the years after the war, many more in the wizarding world required dragon-tear as part of a widespread rise in the popularity of spells. On its own (not including the change in the population of dragons), how has this change in popularity of spells changed the market for dragon-tear?

- e) The demand curve: *shifted in, stayed the same, shifted out* f) The supply curve: *shifted in, stayed the same, shifted out*

Without using numbers, how have these two changes together impacted the dragon-tear market equilibrium?

- g) Equilibrium price has: *increased, stayed the same, decreased, indeterminate*
- h) Equilibrium quantity has: *increased, stayed the same, decreased, indeterminate*

The supply and demand equations for dragon-tear after the post-war changes above are known to be the following:

$$P_d = 69 - \frac{1}{10}Q_d \quad P_s = 15 + \frac{1}{5}Q_s$$

Use a graph to discuss how consumer surplus and producer surplus changed after the war.

- i) Consumer surplus after the war: _____ Change in consumer surplus after the war: _____
- j) Producer surplus after the war: _____ Change in producer surplus after the war: _____

Q6 | Price Elasticity (B5.2)

- a) Use the midpoint method to find the elasticity of demand when the price changes from 10 to 20. _____

- b) Use the midpoint method to find the elasticity of demand when the price changes from 40 to 50. _____
- c) Use the midpoint method to find the elasticity of supply when the price changes from 20 to 40. _____
- d) Which of a)–c) is elastic, unit elastic, or inelastic? _____